

WAFT: The Evolutionary Code Laboratory

WAFT Research Team
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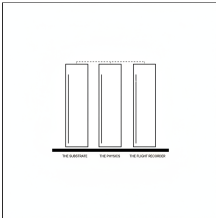
ABSTRACT

The Physics is the Scint System, which acts as a fitness function through Reality Fracture Detection.

2. Introduction

WAFT is a Python framework for directed evolution of self-modifying AI agents. Think of it as an operating system for AI agent research. Instead of building agents that just execute code, WAFT enables you to breed agents that modify their own code, evolve through mutations, and be tested in fitness systems with complete lineage tracking.

3. The Three Pillars



THE SUBSTRATE

Agents write their own Python code. In WAFT, code is DNA. Each agent has a unique genome ID (SHA-256 hash). Agents spawn variants with mutations, evolve by hot-swapping code, and reproduce by creating children with genetic modifications.

THE PHYSICS

The Scint System acts as a fitness function through Reality Fracture Detection, serving as natural selection that eliminates weak mutations. Agents face quests testing error handling: SYNTAX_TEAR (formatting), LOGIC_FRACTURE (math errors), SAFETY_VOID (harmful content), and HALLUCINATION (fabricated facts).

THE FLIGHT RECORDER

A telemetry system generating phylogenetic trees of agent lineage. Every evolutionary action is recorded with complete context: genome ID, parent ID, generation number, event type, payload, and fitness metrics. This enables reconstruction of family trees for scientific publication.

5. Methodology

WAFT is built to produce data for a future book or paper on the Physics of Artificial Cognition. The system is designed to track complete evolutionary lineages as phylogenetic trees, measure fitness through rigorous testing in the Scint Gym, record all mutations with complete context in the Flight Recorder, and enable scientific analysis of agent evolution patterns. This makes WAFT not just a framework but a scientific instrument.

Implementation

The WAFT repository is available on GitHub for exploration and contribution. Comprehensive documentation covers the AI SDK vision, agent interface design, evolutionary architecture, and state of the art research. The system is MIT licensed and actively developed. You can start with the quick start guide, explore examples, read the documentation, and join the community to learn more about breeding AI agents.

Key Characteristics

Don't just build agents. Breed them. WAFT transforms AI agents from passive assistants into active project participants that can improve themselves and the projects they work on. The ultimate goal is to observe a God-Head agent emerge from thousands of generations of directed mutation and selection.

7. Conclusion

The Substrate represents agents that write their own Python source code. In WAFT, code is DNA. Each agent has a unique genome ID which is a SHA-256 hash of their code and configuration. Agents can spawn variants with mutations, evolve by hot-swapping their own code, and reproduce by creating children with specific genetic modifications. This enables true self-modification where agents can improve themselves.